ECA1003 - MICROPROCESSOR AND MICROCONTROLLERS

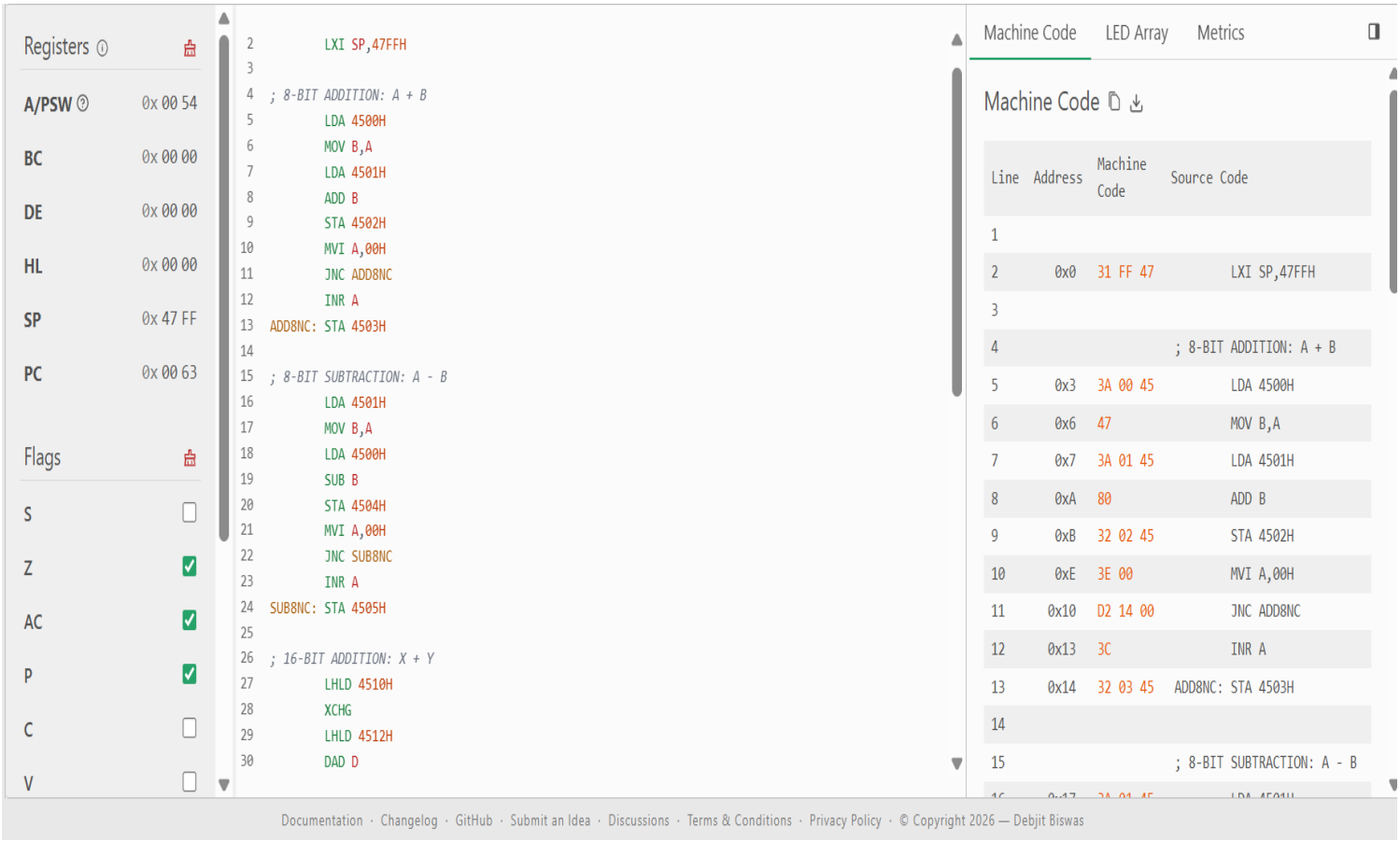
(1) Develop and execute an 8-bit Microprocessor program to add and subtract two 8-bit numbers A = 5AH, B = 27H and two 16-bit numbers X = 1234H, Y = 4321H stored in memory. Store all results in consecutive memory locations, record the carry/borrow for each operation and validate the results by manual hexadecimal calculation. Repeat for A = E8H, B = 9CH, X = F234H, Y = 89ABH.

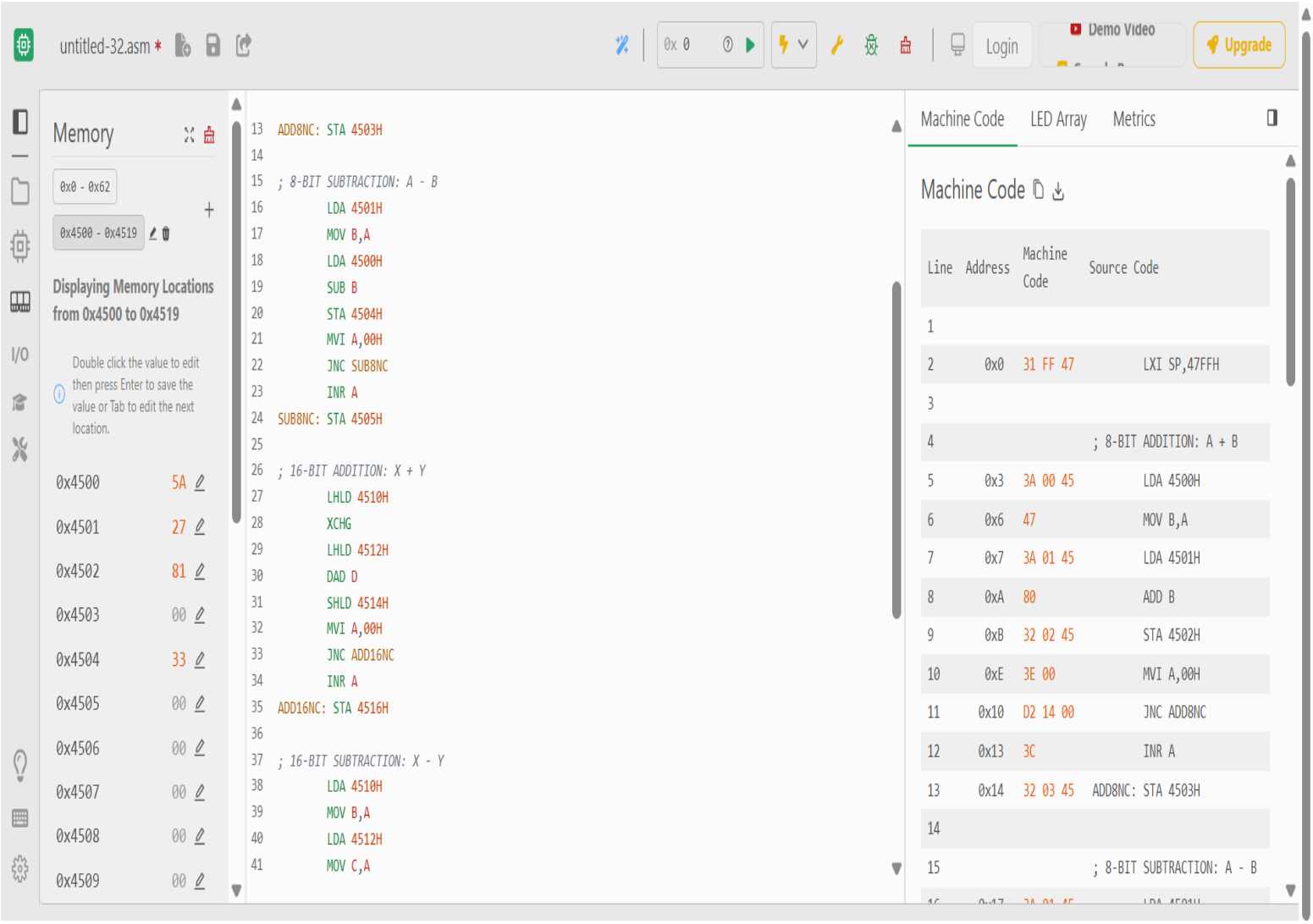
- PROGRAM CODE

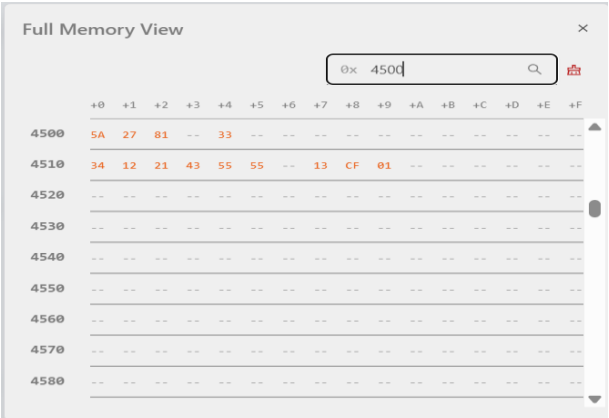
```
LXI SP,47FFH ; 8-BIT ADDITION: A + B
LDA 4500H
MOV B,A
LDA 4501H
ADD B
STA 4502H
MVI A,00H
JNC ADD8NC
INR A
ADD8NC: STA 4503H ; 8-BIT SUBTRACTION: A - B
LDA 4501H
MOV B,A
LDA 4500H
SUB B
STA 4504H
MVI A,00H
JNC SUB8NC
INR A
SUB8NC: STA 4505H ; 16-BIT ADDITION: X + Y
LHLD 4510H
XCHG
LHLD 4512H
DAD D
SHLD 4514H
MVI A,00H
JNC ADD16NC
INR A
ADD16NC: STA 4516H ; 16-BIT SUBTRACTION: X - Y
LDA 4510H
MOV B,A
LDA 4512H
MOV C,A
MOV A,B
SUB C
STA 4517H
LDA 4511H
MOV B,A
```

LDA 4513H
MOV C,A
MOV A,B
SBB C
STA 4518H
MVI A,00H
JNC SUB16NC
INR A
SUB16NC: STA 4519H
HLT

- SET 1







Set 1: A = 5AH, B = 27H, X = 1234H, Y = 4321H

Operation	Result	Carry/Borrow
8-bit addition: 5AH + 27H	81H	Carry = `00H`
8-bit subtraction: 5AH – 27H	33H	Borrow = `00H`
16-bit addition: 1234H + 4321H	5555H	Carry = `00H`
16-bit subtraction: 1234H – 4321H	CF13H	Borrow = `01H`

- SET 2

Displaying Memory Locations from 0x4500 to 0x4519

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

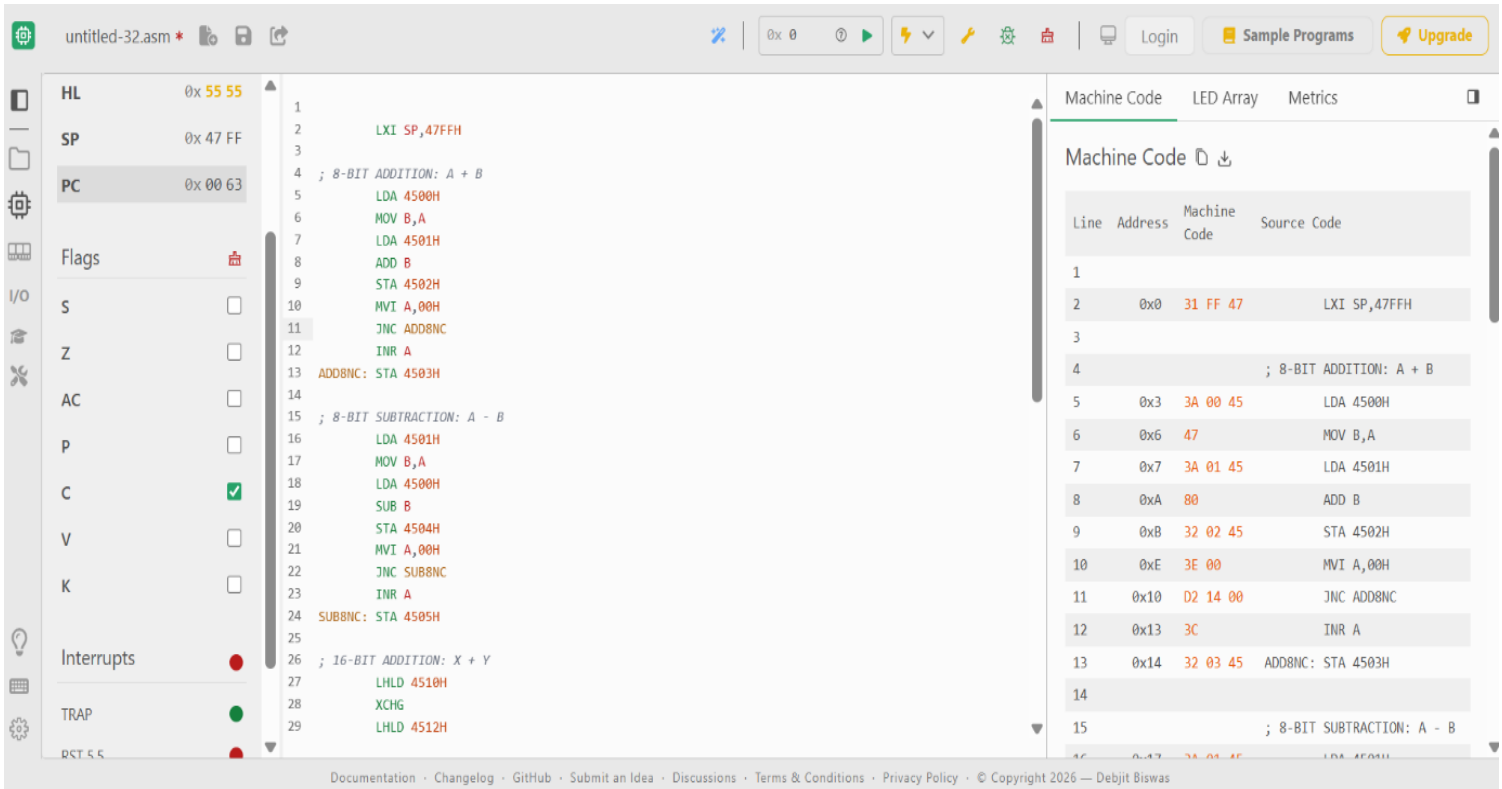
0x4500	E8
0x4501	9C
0x4502	84
0x4503	01
0x4504	4C
0x4505	00
0x4506	00
0x4507	00
0x4508	00
0x4509	00
0x450A	00
0x450B	00
0x450C	00
0x450D	00

```
1       LXI SP,47FFH
2
3  ; 8-BIT ADDITION: A + B
4       LDA 4500H
5       MOV B,A
6       LDA 4501H
7       ADD B
8       STA 4502H
9       MVI A,00H
10      JNC ADD8NC
11      INR A
12 ADD8NC: STA 4503H
13
14 ; 8-BIT SUBTRACTION: A - B
15      LDA 4501H
16      MOV B,A
17      LDA 4500H
18      SUB B
19      STA 4504H
20      MVI A,00H
21      JNC SUB8NC
22      INR A
23 SUB8NC: STA 4505H
24
25 ; 16-BIT ADDITION: X + Y
26      LHLD 4510H
27      XCHG
28      LHLD 4512H
29      DAD D
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2			
3			; 8-BIT ADDITION: A + B
4	0x3	3A 00 45	LDA 4500H
5	0x6	47	MOV B,A
6	0x7	3A 01 45	LDA 4501H
7	0xA	80	ADD B
8	0xB	32 02 45	STA 4502H
9	0xE	3E 00	MVI A,00H
10	0x10	D2 14 00	JNC ADD8NC
11	0x13	3C	INR A
12	0x14	32 03 45	ADD8NC: STA 4503H
13			
14			; 8-BIT SUBTRACTION: A - B
15	0x17	3A 01 45	LDA 4501H



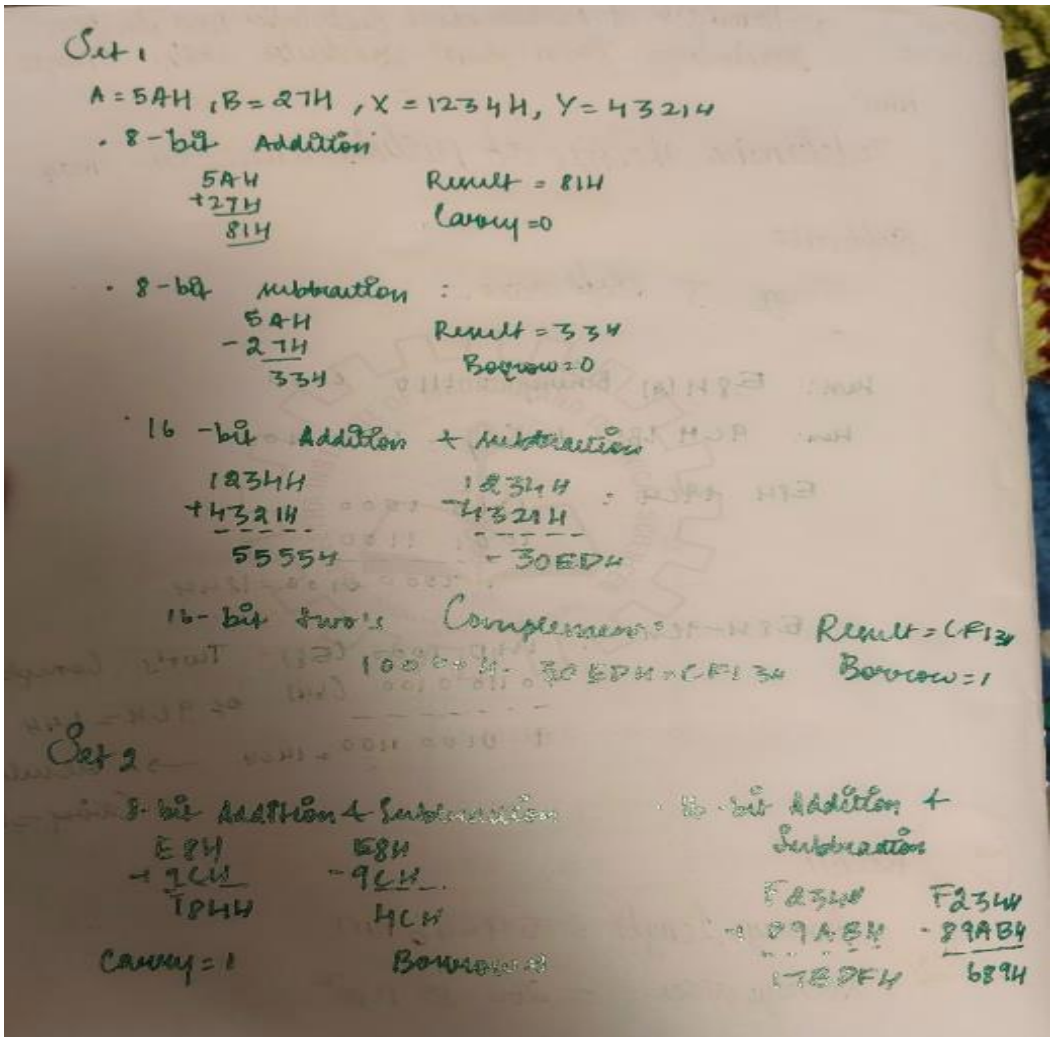
Full Memory View

0x 4500

	+0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+A	+B	+C	+D	+E	+F
4500	E8	9C	84	01	4C	--	--	--	--	--	--	--	--	--	--	--
4510	34	F2	AB	89	DF	7B	01	89	68	--	--	--	--	--	--	--
4520	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4530	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4540	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4550	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4560	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4570	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
4580	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Set 2: A = E8H, B = 9CH, X = F234H, Y = 89ABH

Operation	Result	Carry/Borrow
8-bit addition: E8H + 9CH	84H	Carry = `01H`
8-bit subtraction: E8H − 9CH	4CH	Borrow = `00H`
16-bit addition: F234H + 89ABH	7BDFH	Carry = `01H`
16-bit subtraction: F234H − 89ABH	0689H	Borrow = `00H`



2) For A = E8H, B = 9CH record S, Z, AC, P and CY after addition and subtraction using PUSH PSW, find signed overflow manually, identify the cases where CY and overflow differ, and justify each flag using the signed and unsigned interpretation of the operands.

- PROGRAM CODE

```
LXI  SP,47FFH
LXI  H,4500H
MOV  A,M
INX  H
MOV  B,M
ADD  B
PUSH PSW
STA  4502H
MVI  A,00H
ACI  00H
STA  4503H
POP  H
MOV  A,L
STA  4504H
LDA  4500H
SUB  B
PUSH PSW
STA  4505H
MVI  A,00H
ACI  00H
STA  4506H
POP  H
MOV  A,L
STA  4507H
HLT
```

Memory

0x0 - 0x2E | 0x47FE | 0x4500 - 0x4507

Displaying Memory Locations from 0x4500 to 0x4507

Double click the value to edit then press Enter to save the value or Tab to edit the next location.

0x4500	E8
0x4501	9C
0x4502	84
0x4503	01
0x4504	B5
0x4505	4C
0x4506	00
0x4507	00

```
1  LXI  SP,47FFH
2  LXI  H,4500H
3  MOV  A,M
4  INX  H
5  MOV  B,M
6  ADD  B
7  PUSH PSW
8  STA  4502H
9  MVI  A,00H
10 ACI  00H
11 STA  4503H
12 POP  H
13 MOV  A,L
14 STA  4504H
15 LDA  4500H
16 SUB  B
17 PUSH PSW
18 STA  4505H
19 MVI  A,00H
20 ACI  00H
21 STA  4506H
22 POP  H
23 MOV  A,L
24 STA  4507H
25 HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2	0x3	21 00 45	LXI H,4500H
3	0x6	7E	MOV A,M
4	0x7	23	INX H
5	0x8	46	MOV B,M
6	0x9	80	ADD B
7	0xA	F5	PUSH PSW
8	0xB	32 02 45	STA 4502H
9	0xE	3E 00	MVI A,00H
10	0x10	CE 00	ACI 00H
11	0x12	32 03 45	STA 4503H
12	0x15	E1	POP H
13	0x16	7D	MOV A,L
14	0x17	32 04 45	STA 4504H
15	0x1A	3A 00 45	LDA 4500H
16	0x1D	90	SUB B

Registers

A/PSW	0x 00 44
BC	0x 9C 00
DE	0x 00 00
HL	0x 4C 00
SP	0x 47 FF
PC	0x 00 2F

Flags

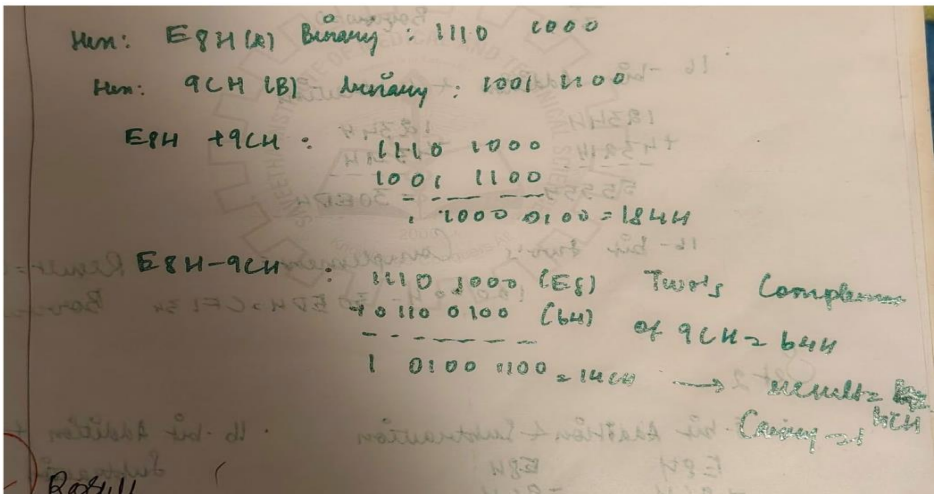
S	☐
Z	☑
AC	☐
P	☑
C	☐
V	☐


```
1  LXI  SP,47FFH
2  LXI  H,4500H
3  MOV  A,M
4  INX  H
5  MOV  B,M
6  ADD  B
7  PUSH PSW
8  STA  4502H
9  MVI  A,00H
10 ACI  00H
11 STA  4503H
12 POP  H
13 MOV  A,L
14 STA  4504H
15 LDA  4500H
16 SUB  B
17 PUSH PSW
18 STA  4505H
19 MVI  A,00H
20 ACI  00H
21 STA  4506H
22 POP  H
23 MOV  A,L
24 STA  4507H
25 HLT
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1	0x0	31 FF 47	LXI SP,47FFH
2	0x3	21 00 45	LXI H,4500H
3	0x6	7E	MOV A,M
4	0x7	23	INX H
5	0x8	46	MOV B,M
6	0x9	80	ADD B
7	0xA	F5	PUSH PSW
8	0xB	32 02 45	STA 4502H
9	0xE	3E 00	MVI A,00H
10	0x10	CE 00	ACI 00H
11	0x12	32 03 45	STA 4503H
12	0x15	E1	POP H
13	0x16	7D	MOV A,L
14	0x17	32 04 45	STA 4504H
15	0x1A	3A 00 45	LDA 4500H
16	0x1D	90	SUB B



3. Develop an 8-bit Microprocessor program to perform 8-bit multiplication and division (A = 14H, B = 06H) and 16-bit multiplication and division (X = 1357H, Y = 0025H) using repeated addition and repeated subtraction. Store the product, quotient and remainder separately and compare the number of loop iterations in each case. Repeat for A = F2H,B = 0AH, X = 9A23H, Y = 0011H.

- PROGRAM CODE :

```
; 8-BIT MULTIPLICATION: 14H * 06H
LXI H,0000H
MVI A,14H
MVI B,06H
MOV C,B
MOV E,A
MVI D,00H
MVI A,00H
MUL8: ADD E
JNC M8NEXT
INR D
M8NEXT: DCR C
JNZ MUL8
MOV L,A
MOV H,D
SHLD 4500H

; 8-BIT DIVISION: 14H / 06H
MVI A,14H
MVI B,06H
MVI C,00H
DIV8: CMP B
```

```
JC D8DONE
SUB B
INR C
JMP DIV8
D8DONE: STA 4502H
MOV A,C
STA 4503H

; 16-BIT MULTIPLICATION: 1357H * 0025H
LXI H,0000H
LXI D,1357H
LXI B,0025H
MUL16: DAD D
DCX B
MOV A,B
ORA C
JNZ MUL16
SHLD 4504H

; 16-BIT DIVISION: 1357H / 0025H
LXI H,1357H
LXI D,0025H
LXI B,0000H
DIV16: MOV A,H
CMP D
JC D16DONE
JNZ D16SUB
MOV A,L
```

CMP E

JC D16DONE

D16SUB: MOV A,L

SUB E

MOV L,A

MOV A,H

SBB D

MOV H,A

INX B

JMP DIV16

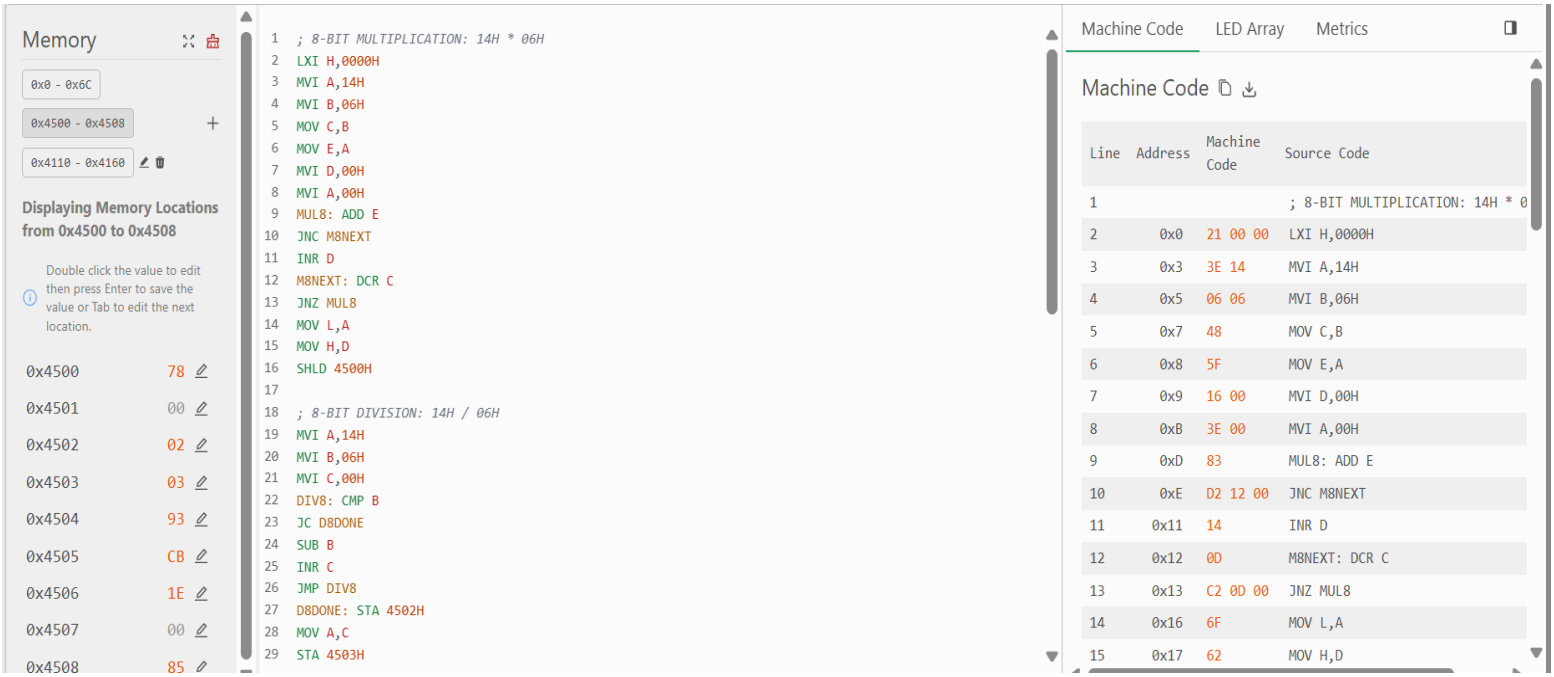
D16DONE: SHLD 4506H

MOV H,B

MOV L,C

SHLD 4508H

HLT



Registers

A/PSW	0x 1E B5
BC	0x 00 85
DE	0x 00 25
HL	0x 00 85
SP	0x FF FF
PC	0x 00 6D

Flags

S	☑
Z	☐
AC	☑
P	☑
C	☑
V	☐

```
; 8-BIT MULTIPLICATION: 14H * 06H
LXI H,0000H
MVI A,14H
MVI B,06H
MOV C,B
MOV E,A
MVI D,00H
MVI A,00H
MUL8: ADD E
JNC M8NEXT
INR D
M8NEXT: DCR C
JNZ MUL8
MOV L,A
MOV H,D
SHLD 4500H

; 8-BIT DIVISION: 14H / 06H
MVI A,14H
MVI B,06H
MVI C,00H
DIV8: CMP B
JC D8DONE
SUB B
INR C
JMP DIV8
D8DONE: STA 4502H
MOV A,C
STA 4503H
```

Machine Code | LED Array | Metrics

Machine Code

Line	Address	Machine Code	Source Code
1			; 8-BIT MULTIPLICATION: 14H * 0
2	0x0	21 00 00	LXI H,0000H
3	0x3	3E 14	MVI A,14H
4	0x5	06 06	MVI B,06H
5	0x7	48	MOV C,B
6	0x8	5F	MOV E,A
7	0x9	16 00	MVI D,00H
8	0xB	3E 00	MVI A,00H
9	0xD	83	MUL8: ADD E
10	0xE	D2 12 00	JNC M8NEXT
11	0x11	14	INR D
12	0x12	0D	M8NEXT: DCR C
13	0x13	C2 0D 00	JNZ MUL8
14	0x16	6F	MOV L,A
15	0x17	62	MOV H,D

